

What is JSON

Q1, what is JSON, where it is used

- text format for storing and exchanging data.
- It's light-weight, human-readable and language-independent (through derived from JS)
- JSON - Javascript Object Notation
- It is string format looks alike JS objects.
- support by almost every language
- ^{for} data exchange b/w servers & applications

Mostly used for:

- API responses
- Configuration files
- Data Storage
- Web communication
- Saving User preference

Q3: Diff B/w JSON String
JS object

JSON String

```
{ "name": "John",  
  "age": 25 }
```

- Keys must be in double quotes

- Cannot contain function
- Can't contain undefined
- Text format for transmission

JavaScript Object

```
{ name: "John",  
  age: 25 }
```

- Keys can be unquoted.

- Can contain functions
- Can contain undefined
- Lives in memory

JSON Syntax Rules

- Keys must be strings in double quotes.
- Values can be: string, number, number, boolean, array, object, null
- No trailing commas
- No comment allowed
- can have nested ~~keys~~ objects

```
{  
  "name": "John",  
  "age": 25,  
  "city": null,  
  "hobbies": ["reading", "coding"],  
  "address": { → nested object  
    "street": "123 Main St",  
    "city": "New York"  
  }  
}
```


JSON.stringify()

→ JSON object to JSON string

```
let student = {
```

```
  name: "Alice",
```

```
  age: 20,
```

```
};
```

```
let jsonString = JSON.stringify(student);
```

```
console.log(typeof jsonString); // String
```

JSON.parse()

→ JSON string to JS Object

```
let jsonString = '{"name": "Alice", "age": 20}';
```

```
let student = JSON.parse(jsonString);
```

```
console.log(student.name);
```

```
console.log(typeof student); // Object
```


JSON Object of a Student

```
{  
  "name" : "Shumaila",  
  "age" : 23,  
  "subjects" : ["Maths", "Physics", "English"]  
}
```

What happens if JSON file uses single quoted instead of double quotes?

It becomes invalid JSON and `JSON.parse()` will throw a syntax ERROR.

What is API?

An API is a set of protocols and rules that allows different software applications to communicate with each other.

Types of API:

Rest API

- Uses HTTP methods (GET, POST, PUT, DELETE)
- Works with URLs and endpoints
- Return data in JSON format
- Stateless (each request is independent)

REST API endpoints

GET /users

— Get all users

GET /users/123

— Get user with ID 123

POST /users

— Create new user

PUT /users/123

— Update user 123

DELETE /users/123

— Delete user 123

Status Code

what does 404
500 mean

Success Codes

- 200 - OK (request succeeded)
- 201 - Created (new resource created)

Client Error Codes

- 400 - Bad Request (invalid data sent)
- 401 - Unauthorized (login required)
- 404 - Not found (resource doesn't exist)

Server Error Codes

- 500 - Internal Server Error (something wrong with server)
- 503 - Server Unavailable
(server temporarily down)

fetch() for API call

```
fetch('https://...../users')
```

- then (response $\Rightarrow$ response.json())
- then (data $\Rightarrow$ console.log(data))
- catch (error $\Rightarrow$ console.error(error))

With Async/await & Error Handling

```
async function fetchUser() {
```

```
  try {
```

```
    let response = await fetch  
      ("https://.....data");
```

```
    if (!response.ok) {
```

```
      throw new Error(response.status);
```

```
    }
```

```
    let data = await response.json();
```

```
    return data;
```

```
  } catch (error) {
```

```
    console.error(error.message);
```

```
    throw error;
```

```
  }  
});
```

```
}
```


What is the role of API?

- for frontend application to get data from backend servers.
- Integrate with third-party services
- mobile apps to sync with web services
- Micro services to communicate with each other